

Patent Report: The Number System Repository

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Abstract

The Number System Repository is a web app that comprises numeral converters between Arabic numerals and different languages across minority cultures. It utilises a set of Python and Streamlit algorithms and coded dictionaries to create converters that directly map between numeric words of English/Arabic and those of other languages, including Chinese, Tibetan, Thai, Hindi, Bengali, Tamil, Yoruba, Igbo, Swahili, Roman, Greek, Inuktitut, Yupik, Quechua, Basque, and more. Aside from the converters, the app also incorporates detailed pages explaining the linguistic nuances of each language. It also contains a holistic database for Linguistics Olympiad problems under the Number Systems category, collating problems across various national and international olympiads such as IOL, PLO, UKLO, NACLO, and many more exam papers into one repository. Together, the Number System Repository is a platform that combines numeral converters, linguistic conceptual explanations, and Linguistic Olympiad problems all in one place, benefiting students preparing for linguistic olympiads and enthusiasts in number systems.

Background and Field of Invention

Target Audience & Market

The primary target audience of the app is students preparing for Linguistics Olympiads: these exams frequently involve number system problems, where a combination of theoretical knowledge and creative thinking is required to solve the numeric rules of a certain numeral system. However, traditionally, these problems have consistently had one of the lowest solve rates due to their complexity. To reduce this gap, I introduced this app as a resource database for students to both understand different number systems and conduct practice on problems.

The secondary target audience of the app is linguistic enthusiasts and those passionate about the conservation of languages, especially number systems. This repository presents linguistic theory pages that document and discuss the important nuances of how the rules of number systems change across different cultures, an often overlooked aspect by the common public and even among linguistic preservation efforts themselves, which often focuses on morphology, semantics, and other more mainstream studies. This app works to mitigate this gap in linguistic conservation and help number system enthusiasts.

Current Landscape

Certain sites exist that have a selected number of numeral converters to different languages, but they are all limited in their scope. There is a lack of a collated numeral converter system that allows both

directions of conversion freely, offers templates for users to enter numbers and obtain an immediate output in the other language, and combines theory with converters. These are functions that our solution possesses.

Furthermore, Linguistics Olympiad problems are heavily fragmented across different exam papers from different competitions, and do not have a holistic repository in which the users can sort problems based on property tags, click on problems and solve them, and then access the solutions through a button. This is the first problem repository app that collates different Linguistics Olympiad problems and activates them within a difficulty-sorted, specialized, problem-solution framework that the user can engage with.

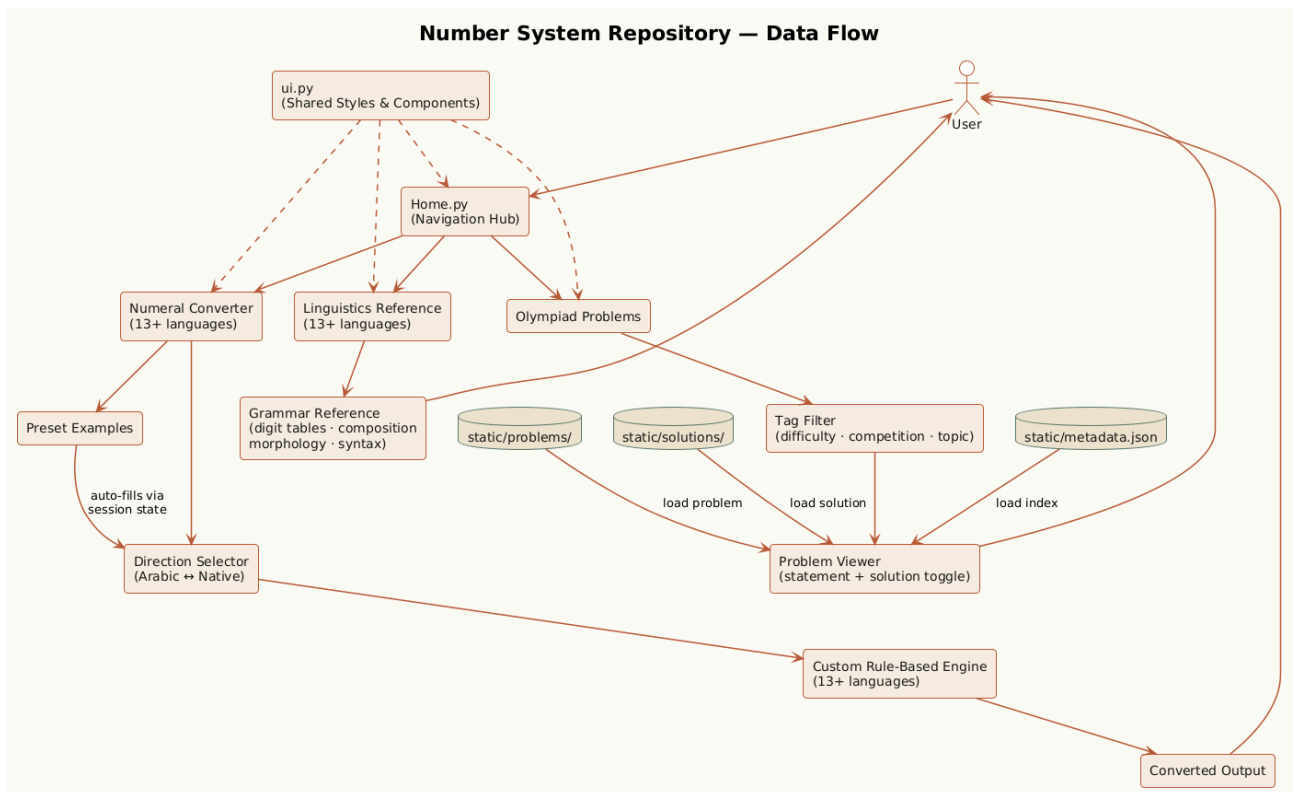
Summary of Invention

Number System Repository is designed to facilitate the understanding, conversion, and study of numeral systems. It is comprised of three major functionalities: numeral converters that convert between arabic numerals and the scripts of other languages, linguistic explanation pages that documents the structural grammar of each system, and a searchable, tagged database of Linguistics Olympiad problems drawn from competitions including IOL, UKLO, NACLO, PLO, APLO, and HKLO. It currently contains 15 languages for the numeral converters and linguistic pages from 6 language families, as well as 13+ Olympiad Problems for exploration and practice.

Detailed Description

Key Components

Data Flow Diagram:



Software Specifications:

The converters are implemented from first principles in pure Python through StreamLit with no external numeral-conversion dependencies, and all code for converters is original. Standard Python modules such as `son`, `pathlib`, `os`, `re`, `unicodedata` are utilized. The problem repository uses a JSON flat file (`static/metadata.json`) as its backing store, with problem and solution files stored locally in `static/problems/` and `static/solutions/` respectively.

Design Specifications:

The application's design is implemented in `ui.py` and applied globally via `apply_global_styles()`. The type system uses two typeface families loaded from Google Fonts, *Crimson Pro* (used for body text, display headings, and result values) and *DM Sans* (used for UI labels, section markers, and navigation).

In terms of color and style, the app incorporates:

- A color palette of four ink colors (`--ink`, `--ink-soft`, `--ink-muted`, `--ink-faint`)
- Parchment surfaces (`--parchment`, `--parchment-2`, `--parchment-3`) for backgrounds and cards
- A terracotta accent (`--accent`, `--accent-light`, `--accent-pale`) for interactive elements, highlights, and callouts
- A secondary teal accent (`--teal`, `--teal-pale`) for result cards and informational panels.

All design tokens are declared as CSS custom properties on `:root`.

The layout of the pages are highly uniform. Converter pages are structured in five sections: masthead, direction selector (a styled radio toggle between the two conversion directions), preset examples (an

eight-button grid), conversion interface (an input card, text field, and result or error card), and a caption-and-navigation footer. Linguistics pages are structured in four to six sections: masthead, system overview with a property grid and a list of key facts, tables and examples describing numerals and digits, formula cards and example blocks for linguistic rules, and optionally syntax-and-morphology and ordinal-formation sections, finally followed by a navigation footer.

Features and Functionality

Converter Engines:

Each language presented in the app contains a Numeral Converter page, where a converter engine is presented. The user is able to choose the direction of conversion, either from Arabic to the target language or vice versa. Then, the user can enter input in two ways: either enter the number in the input language or choose from a set of presets. The output is the equivalent of the entered number in the target language, with base conversions, number-to-text translations, and special phenomena such as overcounting taken into account.

For certain languages, such as Tamil, the output translations for the Arabic to Tamil direction are presented in multiple forms: digit form, word form, and romanized form.

Linguistic Pages:

Each language possesses a linguistic page that explains information about that language. The section split in each page varies, but they usually contain the sections “System Overview”, “Digits & Bases”, and “Compositional Rules”, followed by varying sections of “Script & Pronunciation”, “Syntax & Morphology”, “Noun-Class Agreement”, “Special Conventions”, and more. The information is collated and presented in tabular and bullet point formats, designed to help users understand the linguistic nuances behind each number system they are exploring.

Repository:

The repository contains a series of olympiad problems extracted from different Linguistic Olympiad exams: IOL, PLO, UKLO, NACLO, and others. Each problem is presented on the Olympiad Repository page as their names, with a “View” button next to each name. Clicking on the “View” button allows access to the problem by redirecting the user to a new page with the problem presented as an image or a PDF, and clicking on the “Show Solution” button allows access to the solution image on the same page.

The user can search through the different problems based on the tags: "Chaos & Order", "Table", "Equations", "Special Format", "Time", "Overcounting", "Subtractive systems", "Body Part Counting", "Morphological Changes", "Easy", "Medium", "Hard", "APLO", "IOL", "UKLO", "HKLO", "PLO", "NACLO".

Claims

1. Bidirectional conversion between Arabic numerals and numeral systems of 19 different languages.

The Python code for each numeral converter is original and created by Yi Zou. The system handles a wide range of linguistically complex phenomena that standard numeral conversion tools do not address, including subtractive numeral constructions (where a value is expressed as a subtraction from a higher base, as in Yoruba), vigesimal and mixed-base systems (such as Basque's base-20 structure and Yupik's vigesimal system with lexicalized sub-pivots at 5, 10, and 15), dual-output systems (such as Igbo, which generates both decimal and vigesimal representations), morphological agreement and inflection (such as Greek's gender-sensitive hundreds forms), multi-script input detection (such as Tamil, which auto-detects whether input is in Tamil script digits, Tamil word forms, or romanized text), and positional irregularities within otherwise regular systems (such as Thai's special forms for 1 and 20).

2. The integration of linguistic explanation modules with numeral converters

Original linguistic explanations of each number system. Incorporation of the linguistic explanation page & the numeral converter page. The integration of these two components creates a unified learning pathway: a user can interact with the converter to explore how numbers are formed, then immediately access the corresponding linguistics page to understand the grammatical rules that govern those formations, both of which are originally developed by Yi Zou.

3. The searchable Linguistics Olympiad problem repository with tagging/filtering features

This invention presents the first known digital repository dedicated specifically to number system problems drawn from linguistics olympiad competitions. No comparable resource currently exists in published or patented form.

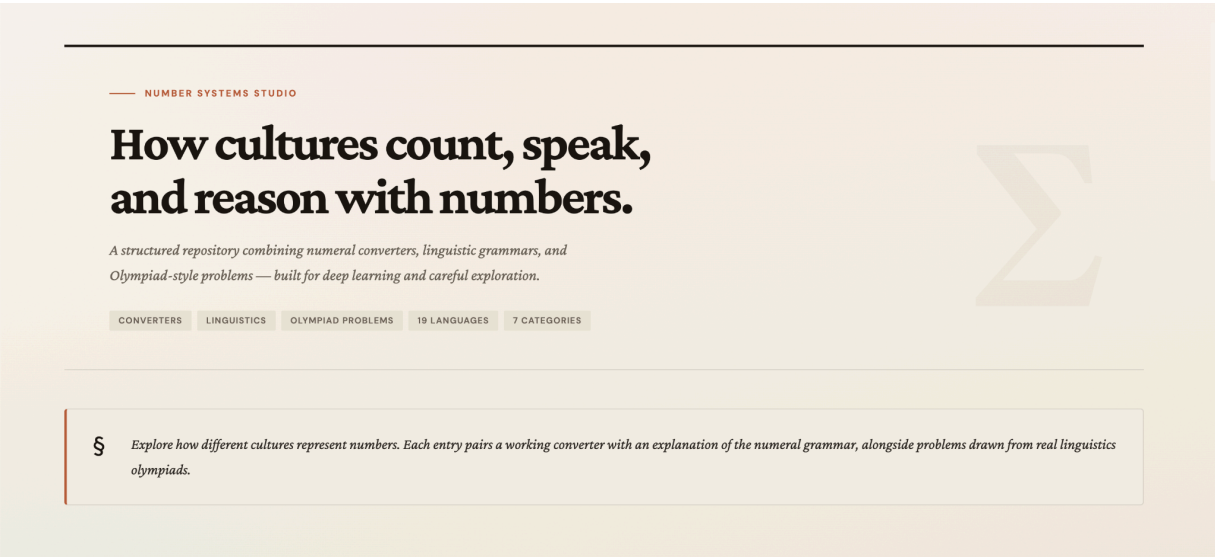
4. The overall interactive educational platform combines converters, linguistic theory pages, and olympiad resources in one unified system

A complete pipeline of each individual element in a complete interface. Each of the three modules independently provides educational value, but their integration within a single platform creates a complete learning pipeline not available in any existing tool.

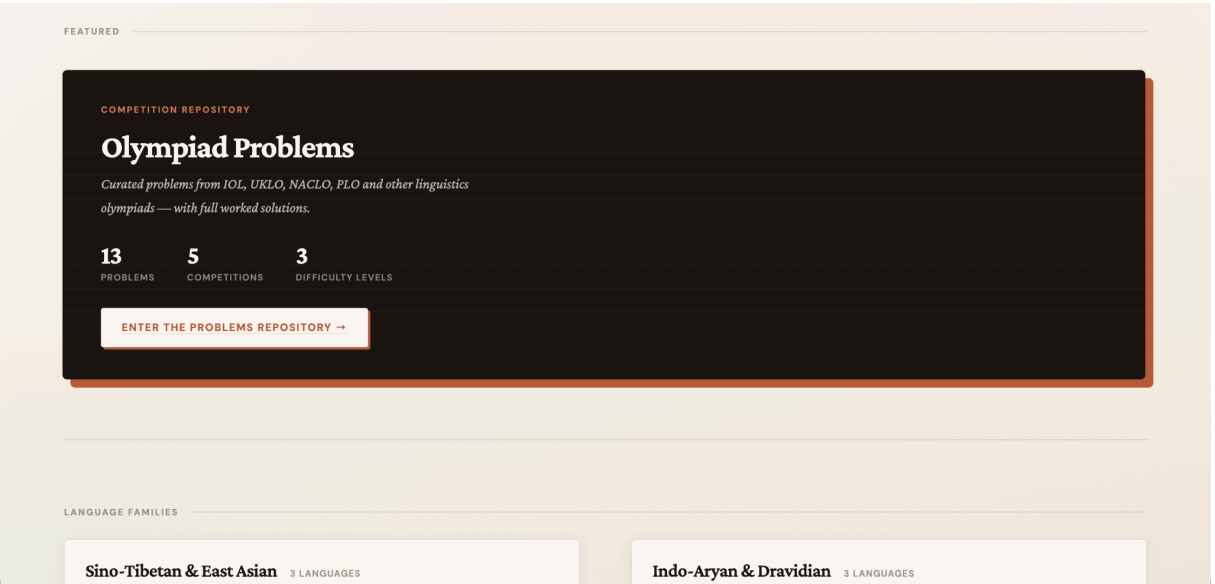
Images

URL: <https://www.numeralatlas.com/>

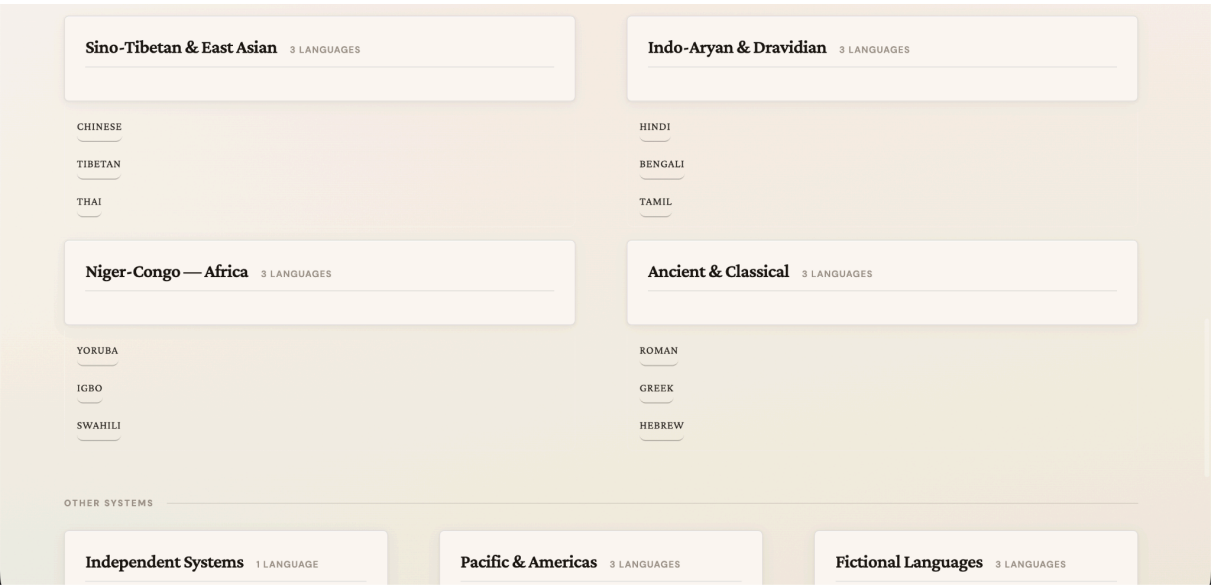
Landing Page:



Number System Repository Entry on the Home Page:



Access to Different Language Families on the Home Page:



Converter Page Example:

NUMERAL CONVERTER

Chinese Numerals

Bidirectional conversion between Arabic numerals and Simplified Chinese, with correct handling of 两, 零 placeholders, and 万/亿 grouping.

CONVERSION DIRECTION

Arabic → Chinese

Chinese → Arabic

PRESET EXAMPLES

CLICK A VALUE TO LOAD IT

01010101000010000012345678

CONVERT

Enter an Arabic numeral

Whole number from 0 to 999,999,999.

e.g. 10100

Implements the Simplified Chinese numeral system. Key rules: 两 replaces 二 before 百/千/万/亿; 零 appears once for any internal zero run; grouping follows 万 (10,000) and 亿 (100,000,000). Range: 0–999,999,999. Converter algorithm by Yi Zou. Grammar detail in the Linguistics section.

CHINESE CONVERTER

CHINESE LINGUISTICS →

← RETURN TO HOME

Part of Linguistics Page:

LINGUISTIC STRUCTURE

Chinese Numerals

DECIMALFULLY MULTIPLICATIVECOMPOSITIONALMORPHOLOGICALLY INVARIANT

10,000-BASED GROUPINGFINANCIAL NUMERAL SYSTEM

SYSTEM OVERVIEW

Structural Properties

KEY FACT

Chinese is the most structurally transparent numeral system in this repository — almost every number above 10 is composed by explicit, fully productive rules. Irregularity is minimal, and the system groups by ten-thousands (10⁴), not thousands.

PRIMARY BASE

10 (decimal)

SYSTEM TYPE

Fully Multiplicative–Additive

GROUPING PIVOT

10⁴ (万 wàn) — not 10³ as in Western systems

HIGHER PIVOT

10⁸ (亿 yì) — groups of ten-thousands above 万

MORPHOPHONOLOGY

Invariant — numerals never inflect

Repository Page:

Number System Problems Repository

Welcome! This repository contains Olympiad-style problems from number systems and linguistics.
Use the **tags** below to filter, then click **View** to see the problem and its solution.

FILTER BY TAGS

Choose options

Umbu-Ungu: IOL 2012 Problem #2

Tags: Table, Hard, IOL, Overcounting

VIEW

Chabu

Tags: Special Format, Other Competitions, Medium

VIEW

Selkup

Tags: Chaos & Order, Overcounting, Subtractive Systems, Medium, Other Competitions

VIEW

Problem Page:

Number System Problems Repository

Welcome! This repository contains Olympiad-style problems from number systems and linguistics.
Use the **tags** below to filter, then click **View** to see the problem and its solution.

← BACK TO LIST

Umbu-Ungu: IOL 2012 Problem #2

Tags: Table, Hard, IOL, Overcounting

Problem

	Umbu-Ungu
10	<i>rureponga talu</i>

	Umbu-Ungu
35	<i>tokapu rureponga yepoko</i>
40	<i>tokapu malapu</i>